Propose, Do Not Certify: Representation Quality and the Judge/Proposer Gap

Physis Research Program

September 25, 2026

Abstract

A representation that is an unreliable judge can still be a useful proposer — but less decisively than a random-split evaluation suggests. On a 582-entry, 57-cell ontology filing task, embedding geometry achieves top-3 containment of the rater-assigned cell at 0.668 under a random train/test split against a matched 1,000-shuffle permuted-label null of 0.159. Under a *sequential* split — first N decisions train, later decisions test, which is what a real user experiences — top-3 falls to 0.541, the learning curve is no longer monotone, and the advantage over the random split is about thirteen points. The proposer's advantage *over chance* survives the protocol change, however: under the sequential split's own 1,000-shuffle permuted null (top-3 0.107) the margin is +0.434, against +0.509 under the random split. The absolute level is inflated by the random split; the effect is not.

We deliberately do *not* subtract the proposer score from the classifier score. Containment in $\{0, 1\}$ and ROC AUC are different metrics on different scales, and their difference has no interpretation; an earlier draft of this paper reported such a "gap" $(0.668 - 0.519)$ and that arithmetic is withdrawn here. Each task is instead reported against its own matched null. Judged that way: as a proposer the representation clears its random-split permuted null by +0.509 top-3, and as a certifier it sits at chance (0.519 against a 0.5 null, $n = 574$). The surviving claim is therefore qualitative and about decision leverage — the representation narrows 57 candidate cells to 3 and places the rater-assigned cell in that set 54–67% of the time, whereas the same representation asked to certify is uninformative — not a numerical gap between two incommensurable scales. One qualification, from a second rater on the same 582 entries: the raters agree on the exact cell for 0.230 of them ($\kappa = 0.202$), and a proposer trained on one rater reaches the other rater's cell in its top-3 at 0.319 and 0.367 against 0.608 and 0.673 for its own rater, while on a single axis most of the signal crosses. The representation carries structure both raters share, but the 57-cell reduction is roughly half as strong for a rater it was not trained on, and what shares is the axis, not the cell.

1 Introduction

Every refutation in the Physis research track refuted unsupervised, zero-shot judgement. COSINE, NAME-ONLY, CONSTRAINT, ANCESTRY and ANCHORVOC are fixed formulas with nothing fitted to a human decision, and on real misfilings they sit at chance (0.508–0.536). The supervised, human-in-the-loop version was never measured — and the question a person filing 582 entries has is not "is the machine right?" but "can it narrow 57 cells to 3 so I choose instead of search?"

The proposition we test is not that geometry can judge — Section 5 shows it cannot — but that it can propose: the representation that cannot judge may still usefully narrow a search. We measure that separately, in each case against the null appropriate to the task, and find the proposer's advantage real under both a random and a sequential split. The certifier was measured once, under

one protocol, and sits at chance there; we did not re-run it sequentially, so we do not claim it is absent under both. We do not report a distance between the two tasks, for the reasons given in Section 7.

2 Method

Ground truth is the rater's *revised* cell in `refiling_gate0.tsv`: 582 blind re-filings, 57 distinct target cells. The 76 non-blind entries are excluded. Vectors are the shipped bge-base-en-v1.5 768-d from `corpus.json`; no inference is run.

Nearest-centroid over cells built from the decisions already made, scored as top-k containment of the rater-assigned cell, on a learning curve over how many decisions the rater has made. Five shuffles per point.

3 Results

Decisions seen	Top-1	Top-3	Top-5
58	0.211	0.385	0.465
145	0.307	0.508	0.591
291	0.368	0.595	0.689
436	**0.438**	**0.667**	**0.755**

Table 1: Top-k containment of the rater-assigned cell, under the random split only, by number of decisions seen. This curve does not survive the sequential split (§split protocol): there it falls at the last step, so the earlier "still climbing at 436" reading is withdrawn.

Controls at 436 decisions seen:

Arm	Top-1	Top-3	Top-5
GEOMETRY	**0.438**	**0.667**	**0.755**
PRIOR old-¿new lookup	0.184	0.368	0.474
MAJORITY	0.089	0.255	0.359
PERMUTED (null)	0.036	0.136	0.233
LEN-ONLY	0.010	0.036	0.071

Table 2: Controls at 436 decisions, all arms at five random splits so the comparison is internally consistent. Geometry beats its permuted null by +0.531 top-3, and beats the lookup table by +0.299. The null row is the weakest number here — see the text for the 1,000-shuffle re-estimate.

Geometry beats its permuted null by +0.531 top-3 at this seed count, and beats the lookup table by +0.299. The learning curve appears to be still climbing at 436 under this split — but that claim does not survive the sequential split reported below, where the curve falls at the last step, so we withdraw it.

One row of that table has since been re-estimated. The permuted-null arm averaged only five shuffles, which is too few for a permutation baseline, so it was re-run with 1,000 shuffles at the same split. That raises the null from 0.136 to 0.159 top-3, which narrows the geometry-minus-null gap from +0.531 to +0.509. Geometry under 1,000 splits reads 0.419 / 0.668 / 0.769, against

the 0.438 / 0.667 / 0.755 shown above, so the two estimates agree to within about 0.02 at every k. We report both rather than silently mixing seed counts within one table, and we take +0.509 as the defensible gap.

4 Discussion

Each task against its own null — not against each other. The tempting summary of this result is a single number: proposer top-3 minus certifier AUC. We do not report one, because the two quantities are not commensurable. Top-k containment is a $\{0, 1\}$ hit rate with a finite number of candidate cells; ROC AUC is a rank statistic over a threshold-free continuum. Neither the difference $0.668 - 0.519$ nor the ratio $0.668/0.519$ has an interpretation — changing the candidate-set size from 3 to 10 would move the first term while leaving the second untouched. An earlier draft of this paper reported such a "gap" as the headline finding and computed its shrinkage across splits ($+0.149 \to +0.022$). That arithmetic is withdrawn.

The comparison is instead made within each task. As a proposer, geometry clears its matched permuted-label null by +0.509 top-3 under the random split (0.668 vs. 0.159). As a certifier, it clears its own null — chance, AUC 0.5 — by 0.019, which is not distinguishable from no improvement ($n = 574$). So the asymmetry is real but it is an asymmetry of *which null the task clears*, not a distance between two scores on different rulers.

The sequential protocol now has its own permuted null — 1,000 label shuffles under the same construction, using the sequential split's own training portion:

frac	geometry top-3	null top-3	margin	null top-1	null top-5
0.10	0.345	0.195	+0.150	0.071	0.300
0.25	0.348	0.177	+0.171	0.060	0.275
0.50	0.577	0.166	+0.412	0.055	0.265
0.75	0.541	0.107	+0.434	0.034	0.178

The sequential null is *lower* than the random-split null at the same fraction (0.107 against 0.159 top-3), because a shuffled-label prior built from fewer training entries is weaker. The consequence is that the measured margin under the conservative protocol (+0.434) is close to the random-split margin (+0.509) even though the geometry arm itself drops thirteen points. The proposer's advantage over chance therefore survives the protocol change; what does not survive is its absolute level. We lead with the sequential number and keep the random split as a labelled upper bound.

Put in a common, decision-relevant currency: the proposer takes a 57-cell search and hands a human 3 candidates containing the rater-assigned cell 54–67% of the time (sequential to random), while the certifier, asked to answer the same question without a human in the loop, provides no usable signal at all. That is a statement about human decision leverage, and it is the claim we are prepared to defend.

The magnitude is split-dependent; the effect is not. The proposer's top-3 falls from 0.668 to 0.541 when evaluation becomes sequential — about thirteen points — so the absolute level is inflated by the random split. But the sequential split's own permuted null is also lower (0.107 against 0.159), and the margin over that null barely moves ($+0.509 \to +0.434$). The more realistic protocol lowers the score and the floor together. We therefore lead with the sequential figure and keep the random split as a labelled upper bound. What survives both splits is the ordering: geometry beats its matched null, the lookup prior, the majority baseline and the length-only baseline at every fraction under both protocols, and the certifier sits at chance in its single measurement (0.519, $n = 574$).

The earlier framing, that a number previously filed as a failure (top-1 0.447) becomes useful once the output is three candidates, still holds under both splits: top-3 exceeds top-1 by a factor of about 1.8 even sequentially, and a person choosing among three candidates has less work than a person searching 57 cells.

Supervision is doing the work. 436 decisions is most of a corpus. The honest claim is "geometry converts human decisions into leverage over the remaining ones," not "geometry replaces them." The learning curve is the thing to read: at 58 decisions top-3 is 0.385 against a majority-baseline top-3 of 0.255 (1.5×) and a 3.5× multiple of the 0.11 share held by the single largest cell.

The mode axis is the constraint. Scoring per axis at 436 decisions seen:

Axis	Classes	Top-1	Top-2	Lift over majority
Domain	5	0.626	0.837	3.05×
Mode	14	0.460	0.678	1.97×

Table 3: Per-axis proposer performance. The machine can carry the domain axis and cannot carry the mode axis.

The machine can carry the domain axis and cannot carry the mode axis — 0.837 top-2 over five domains against 0.678 over fourteen modes, and the gap survives normalising for class count. This is the same asymmetry E6/E7 found and explained by population rather than by missing definition, now stated as a division of labour: **the machine proposes the domain, the person picks the mode.**

Whose cells? The proposer against a second rater

Everything above trains on the cells *Claude* assigned and tests containment of cells *Claude* assigned. That cannot separate "the embedding carries semantic structure" from "the embedding memorises one rater's policy." A second rater was therefore run on the same 582 entries — deepseek-chat, blind to the old cell, to Claude's cell and to any geometry proposal, one item at a time — and the four arms crossed on identical splits, each against its own matched null:

Arm	0.10	0.25	0.50	0.75	null @0.75
A→A (Claude → Claude)	0.367	0.501	0.610	0.673	0.159
B→B (DeepSeek → DeepSeek)	0.404	0.492	0.564	0.608	0.206
A→B (train A, test B)	0.226	0.297	0.349	0.367	0.127
B→A (train B, test A)	0.203	0.264	0.302	0.319	0.110

Table 4: Top-3 containment by rater pair, 582 entries, 40 splits + 200 train-label permutations per point. Margins over the matched null at 436 decisions seen: +0.514, +0.402, +0.240, +0.209.

At 436 decisions seen, a proposer trained on one rater puts the other rater's cell in its top-3 at 0.367 and 0.319 against 0.673 and 0.608 for its own rater. Cross-rater transfer clears its own null at every arm and every fraction, so the signal is not purely rater-specific — but at the cell level it keeps only about half the margin, and the DeepSeek-trained arm does not beat the marginals-only baseline at any fraction (0.319 against 0.330 at 436 decisions). Part of that is coverage, not geometry: 0.182 of the test entries in that arm carry a cell the DeepSeek-trained candidates cannot contain, because DeepSeek used 42 cells where Claude used 57 (15 Claude cells are outside DeepSeek's vocabulary,

none the reverse). The two raters agree on the exact cell for 0.230 of entries, $\kappa = 0.202$ (95% CI 0.168–0.235) — slight agreement, so the exact cell is largely a rater's own policy.

The shared part lives at lower resolution, which is where the per-axis split above already said the machine carries the work. Scoring one axis at a time at 436 decisions seen:

Axis	**Arm**	**Top-3**	**Margin**	**Majority**
Domain (5)	A→A	0.948	+0.333	0.698
	B→B	0.922	+0.216	0.839
	A→B	0.809	+0.201	0.698
	B→A	0.790	+0.182	0.839
Mode (14/13)	A→A	0.817	+0.425	0.555
	B→B	0.813	+0.380	0.567
	A→B	0.675	+0.326	0.555
	B→A	0.668	+0.320	0.567

Table 5: One axis at a time, 436 decisions seen. Cross-rater transfer keeps 0.675/0.668 of 0.817/0.813 on the mode axis. The domain-axis B→A majority figure is inherited from a train-side marginal DeepSeek does not share with Claude.

Restricting both raters to the 42 cells they do share (500 entries) leaves the picture where it was (A→B 0.399, B→A 0.363 at 436 decisions seen), so the cell-level attenuation is not an artifact of unequal vocabularies.

The hypothesis this tests — exact cell assignment is rater-dependent, while the geometry carries lower-resolution structure both raters share — survives, and it corrects this paper's own framing rather than confirming it. The 57-cell reduction is a reduction over *one* rater's cells: for a filer whose policy differs from the training rater it is worth roughly half the headline number, and on the exact cell it is worth no more than the trivial baseline of the three most common cells until the rater has made a few hundred decisions. What transfers is the axis, not the cell.

Limits. Two raters, both language models, and the first of them (Claude) also produced the replacement filing. Under "is the machine right" a self-consistent rater is a confound. Under this framing it is the target — the artifact exists to pre-sort for an arbiter — and the second-rater arms above now bound how much of the result is that rater's own policy: about half at the cell level, and much less at the axis level. That bound is not a resolution. Two language models agreeing is agreement between two policies, not evidence about ontology truth, and not evidence about human filers. A human rater is still the experiment that would settle whose policy this is.

What this does not show. (1) Top-1 is not new — 0.438 essentially reproduces E5's 0.447 accuracy on the old labels. (2) It is not certification — nothing here says an entry belongs in the proposed cell. E14–E18 stand untouched. (3) It predicts a policy, not a truth. (4) Both raters were language models, so this measures whether geometry can predict a model's filing policy — and, per the crossed arms, only about half of a policy it was not trained on. It says nothing about ontology truth or about human filers; a human rater remains the experiment that would settle it.

The split protocol, stated precisely

A learning-curve claim is only as good as its split, so we state ours exactly. At each fraction $f \in \{0.10, 0.25, 0.50, 0.75\}$ the 582 entries are shuffled and cut into a training prefix of $\max(2, \lfloor f \cdot 582 \rfloor)$ decisions and the remaining untouched decisions as test. Centroids are built from **train only**; test

entries never contribute to a centroid. Each point is the mean over **five** random splits, and the reported k-seen values are those fractions (58, 145, 291, 436).

Two properties of this protocol are weaknesses, and we name them rather than leave them to be found:

- **The split is random in the main result; a sequential split was run in this revision.** A random split lets a test entry sit between training entries rather than after them, which is not what a real user experiences. We therefore ran the same protocol in file order — first N decisions as training, every later decision as test — and the difference is large. Reporting the sequential results as the more realistic estimate:

frac	train	test	top-1	top-3	top-5	prior-3	maj-3
0.10	58	524	0.195	0.345	0.422	0.250	0.218
0.25	145	437	0.195	0.348	0.430	0.272	0.224
0.50	291	291	0.323	0.577	0.677	0.309	0.230
0.75	436	146	0.308	0.541	0.644	0.267	0.123

  Against the random split at frac 0.75 (top-3 0.667–0.668), the sequential split gives 0.541 — the random split inflates top-3 by roughly +0.13. Two further consequences follow. First, the learning curve is *not* monotone in sequence: it rises from 0.345 to 0.577 and then *falls* to 0.541, so the earlier claim that "the curve is still climbing at 436" does not hold under this split. Second, at small N the geometry arm (0.345, 0.348) is barely distinguishable from the old→new lookup prior (0.250, 0.272), though it still comfortably beats majority (0.218, 0.224).

  Caveat on what this split is: `refiling_gate0.tsv` carries a sequence column but no timestamps, so we cannot confirm that the file order is the rater's adjudication order rather than the corpus order. This is a *sequential* split and we do not claim it is chronological. It bounds the random split's optimism; it does not establish the true temporal curve.

- **The permutation null was re-run with 1,000 shuffles.** The original arm averaged five shuffles, which our own project notes elsewhere call too weak for a permutation baseline. Re-running the null with 1,000 shuffles at the same split (0.75, 436 decisions seen) gives top-3 null 0.159 against the published 0.136. The five-shuffle figure was low by 0.023; we have no interval for it, since the original script reported only the mean over five, so we cannot say whether the two differ significantly — only that the better-estimated null is higher. Corrected numbers at 436 decisions: null top-1 0.048, top-3 0.159, top-5 0.265; geometry top-1 0.419, top-3 0.668, top-5 0.769. The geometry-minus-null gap is therefore +0.371, +0.509 and +0.505 rather than the +0.402, +0.531 and +0.522 implied by the original five-shuffle null. We report the revised figures and the reduced gap. The conclusion is unchanged in direction — geometry clears the permuted null by +0.509 top-3, while the strongest baseline over that same null, the old→new lookup prior, reaches only +0.209 (41% as much) and the length-only baseline does not clear the null at all (−0.123) — but the published null understated the noise, and we prefer to correct it ourselves than have a reviewer find it.

The revised null is stored in `research/e21_permutation_null_1000.json`; the re-run script is `research/papers/e21_perm_null_1000.py`, which imports the original `run_e21.py` so both arms come from the same code path.

Neither the remaining temporal-split gap nor any other weakness invalidates the direction of the result — the geometry arm beats every baseline by a wide margin — but they bound how much precision we claim.

Reproducibility and provenance

The E21 numbers above (top-k containment, the four baselines, the per-axis split) were regenerated from the surviving script and ground truth and are stored in `research/e21_proposer_results.json`; `research/papers/parse_e21.py` rebuilds that file and asserts the two headline figures.

The second-rater arms are stored in `research/e21_cross_rater_results.json`, produced by `research/papers/e21_cross_rater.py` (40 splits and 200 train-label permutations per point for the four main arms; 20 splits and 100 permutations for the shared-vocabulary and per-axis views) and re-checked by `research/papers/check_e21_cross_rater.py`, which also re-derives the inter-rater agreement and κ from the raw files and asserts the invariants (containment never exceeds the candidate-cell ceiling; k-seen splits are 58/145/291/436). Two data facts that the arms depend on: the join to the rater-2 file is on the row index `n`, because entry names repeat in `refiling_gate0.tsv` (573 distinct names over the 585 blind rows, so a name-keyed join silently merges entries); and the arms reproduce the published E21 row exactly on the first five splits, which is the integrity check on the shared pipeline.

The certification figures quoted for context — the 0.519 (COSINE) and 0.536 (CONSTRAINT) AUCs on real misfilings — come from the E15–E18 re-run stored in `research/e15_e18_rerun.json`. Those four experiments had shipped only as printing scripts with no saved output; they have since been re-run from the surviving corpus and ground truth, and reproduced their originally reported values (COSINE 0.882 under random injection and 0.519 on real misfilings; CONSTRAINT 0.612 and 0.536). The E5 figure 0.447 and the E24 figure 0.712 remain *not* independently reproducible: neither experiment has a saved result file in the working repository or the retired archive. They are quoted from the original reports and should be re-run before being relied on.